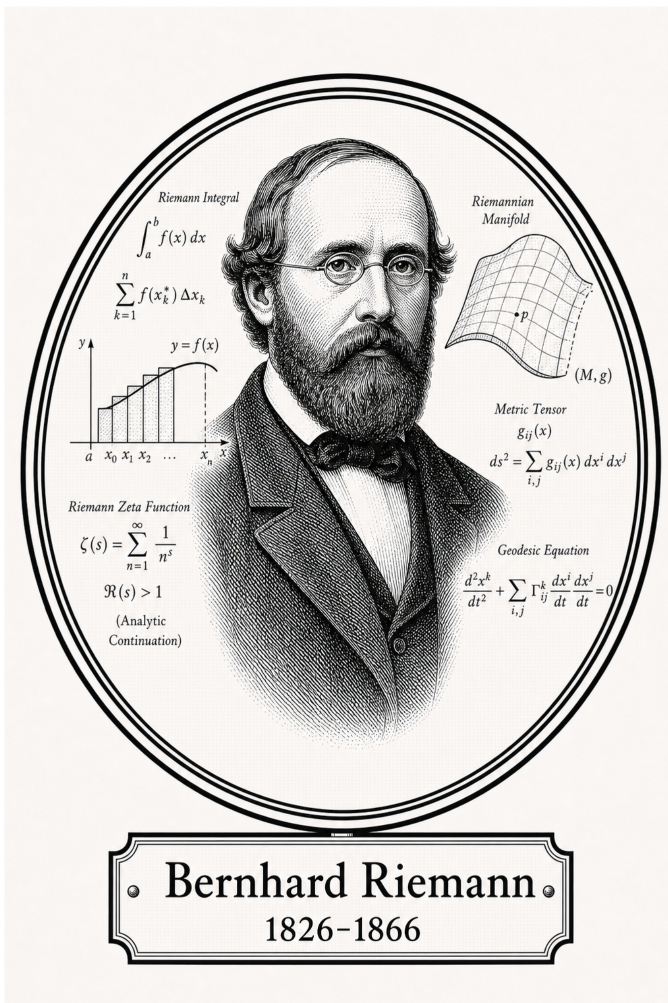
From Cantor to Ito

Classical Analysis

Integration



Bernhard Riemann

1826-1866

Self-Study Notes

William Sollers

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For every reader learning to make mathematics their own.

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Chapter 1

Integration

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1.1 Partitions and Tags

Remark (Reframing note). This chapter is the narrative version of the reference list. The reference orders the definitions correctly, but the real subject is the sequence of defects and patches:

$$\begin{aligned} \text{geometric problem} &\to \text{Cauchy} \to \text{Riemann} \\ &\to \text{Darboux} \to \text{Stieltjes} \\ &\to \text{Henstock–Kurzweil and McShane} \\ &\to \text{Lebesgue.} \end{aligned}$$

Each integral answers a concrete complaint about the one before it. The point is not to memorize several integrals, but to understand why each one had to be invented.

The Geometric Problem

Forget integrals for a moment. The physical question is always the same:

> Given a pointwise rate f, how much total quantity accumulates from a to b?

A speedometer reads $f(t)$; the accumulated quantity is distance. A wire has linear density $f(x)$; the accumulated quantity is mass. A curve $y = f(x)$ over $[a, b]$ asks for signed area. The only available strategy is:

1. Chop $[a, b]$ into thin slabs.
2. Pretend f is constant on each slab.
3. Add the resulting rectangles.

4. Refine the chopping and ask whether the sums settle.

Every construction below makes two choices: which height to use in a slab, and what it means for all these rectangle sums to settle to one number.

Remark (Construction comparison). The constructions differ by the height rule, the ruler, or the fineness rule:

Construction	Height rule	Ruler / fineness
Cauchy	$f(x_{i-1})$, left endpoint	Δx_i
Riemann	$f(\xi_i)$, arbitrary tag	Δx_i
Darboux	$\inf_I f,\ \sup_I f$ on each slab	Δx_i
Riemann–Stieltjes	$f(\xi_i)$, tagged height	$\Delta \alpha_i$
Henstock–Kurzweil	$f(\xi_i),\ \xi_i \in I_i$	$\Delta x_i,\ \delta(\xi_i)$-fine, tag tethered
McShane	$f(t_i),\ t_i$ may float	$\Delta x_i,\ \delta(t_i)$-fine, tag untethered

Henstock–Kurzweil does not change the sampled height or the ordinary ruler; it changes the admissible chopping by replacing a global mesh bound with a local gauge condition. McShane keeps the same gauge idea but frees the tag from the slab, which pulls the theory back toward Lebesgue's absolute notion of size.

Definition (Partition)

Definition 1.1.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\text{Partition}(P, a, b) \iff \exists n \in \mathbb{N}\, \exists x_0, \ldots, x_n \in \mathbb{R}\\ \big(P = \{x_0, \ldots, x_n\} \wedge x_0 = a \wedge x_n = b\big)\\ \wedge \forall i \,\big(1 \le i \le n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition (Subinterval Width)

Definition 1.1.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition (Mesh)

Definition 1.1.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

Remark (Geometric reading). The mesh is the resolution dial. The phrase "as the partition is refined" will usually mean that $\|P\| \to 0$: no slab is allowed to remain fat.

Definition (Tagged Partition)

Definition 1.1.4 (Tagged Partition)**.** A *tagged partition* is a partition $P = \{x_0, \ldots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition (Refinement)

Definition 1.1.5 (Refinement of a Partition)**.** A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma (Common Refinement)

Lemma 1.1.6 (Common Refinement of Two Partitions)**.** *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

Remark (Why the lemma is load-bearing). This small fact is the reason independently chosen partitions can be compared. Whenever a proof says "refine them together," it is using this lemma. It is the structural engine behind Cauchy's existence proof and Darboux's squeeze.

1.2 The Cauchy Integral

Geometric Intuition

Cauchy makes the simplest possible height choice: always sample the left edge of each slab. The result is a staircase of left-anchored rectangles. As the slabs thin out, the question is whether those staircase areas settle.

Remark (Engineering reading). Cauchy's construction is the seed. It answers the geometric problem for the case where the curve is tame enough that left-edge staircases forget their initial sampling bias as the mesh shrinks. The construction is deliberately primitive: it fixes a height rule first and asks for convergence second.

Definition (Cauchy Sum)

Definition 1.2.1 (Left-Endpoint Cauchy Sum)**.** For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\, \Delta x_i.$$

Definition (Cauchy Integral)

Definition 1.2.2 (Cauchy Integral)**.** The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall P\ \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\, dx$.

Proposition (Integral of a Constant)

Proposition 1.2.3 (Cauchy Integral of a Constant). *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b-a)$.*

Theorem (Linearity)

Theorem 1.2.4 (Linearity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem (Monotonicity)

Theorem 1.2.5 (Monotonicity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \leq g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \leq \int_a^b g\,dx.$$

Corollary (Bounds)

Corollary 1.2.6 (Bounds for the Cauchy Integral). *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b-a) \leq \int_a^b f\,dx \leq M(b-a).$$

Theorem (Triangle Inequality)

Theorem 1.2.7 (Triangle Inequality for the Cauchy Integral). *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left|\int_a^b f\,dx\right| \leq \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 1.2.8 (Interval Additivity for the Cauchy Integral). *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition (Oscillation on an Interval)

Definition 1.2.9 (Oscillation on an Interval)**.** For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Remark (Oscillation as the visible error)*.* Figure 1.1 shows the same quantity geometrically: the Darboux gap $U(f,P) - L(f,P)$ is the sum of local oscillations $\omega_f(I_i)$ multiplied by the slab widths Δx_i. The trapezoid and Simpson panels are numerical side branches; the load-bearing panel here is the oscillation identity.

Quadrature & Oscillation

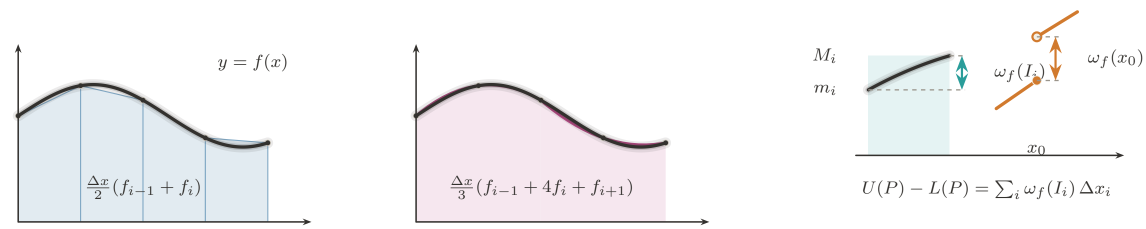

Figure 1. **Trapezoid Rule: Chords: Exact for Affine f**

Figure 2. **Simpson's Rule: Parabolas: Exact to Degree 3**

Figure 3. **Oscillation: The Gap Generator ω_f**

Figure 1.1: Quadrature and oscillation

Theorem (Continuous Implies Cauchy-Integrable)

Theorem 1.2.10 (Continuous Functions Are Cauchy-Integrable)**.** *Every continuous function $f : [a,b] \to \mathbb{R}$ is Cauchy-integrable.*

Remark (Proof skeleton)*.* Continuity on the compact interval gives uniform continuity. Hence short slabs have small oscillation. Given two fine partitions, pass to their common refinement and compare sums slab by slab; each re-sampling error is bounded by $\omega_f(I)$ times the slab width. The sums form a Cauchy net, and completeness of $\mathbb{R}$ supplies the limit.

Remark (What continuity buys)*.* Uniform continuity is the load-bearing ingredient. It converts pointwise control into one global scale: if the slabs are all shorter than that scale, then every slab has small oscillation. Without this global scale, Cauchy's left-endpoint definition has no mechanism for proving that unrelated partitions settle to the same number.

Remark (A first discontinuous edge case)*.* A single jump should still have an area. If

$$f(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c, \end{cases}$$

on $[a, b]$, the only dangerous slab is the one crossing c. Its total possible error is bounded by its width, and that width tends to zero as the mesh tends to zero. This example is not hard; the point is that Cauchy's definition gives no internal test explaining why it is harmless. The proof above proves continuity is sufficient, not that controlled discontinuities are allowed.

Theorem (Tag Independence)

Theorem 1.2.11 (Tag Independence for Continuous Functions)**.** *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

Remark (Bug report)*.* Cauchy has two defects. First, the left endpoint is privileged by convention; tag independence is a theorem only after continuity has already been assumed. Second, the existence proof gives no usable criterion for discontinuous functions. The patch request is to stop privileging the left endpoint and make tag independence part of the definition.

Remark (Patch request)*.* The next definition should not say "sample on the left." It should say: sample anywhere, even adversarially, and require all fine enough choices to agree. Riemann is exactly Cauchy's tag-independence theorem promoted into the definition of integrability.

1.3 The Riemann Integral

Geometric Intuition

Riemann promotes Cauchy's tag-independence theorem into a definition. A function is integrable when every sufficiently fine staircase, even with adversarial tags, is forced near the same number.

Remark (Engineering reading)*.* The sampling convention is no longer chosen by the author of the definition. It is chosen by a universal quantifier. A Riemann-integrable function is one whose accumulated area survives every possible choice of sample heights once the slabs are fine enough.

Definition (Riemann Sum)

Definition 1.3.1 (Riemann Sum)**.** For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition (Riemann Integral)

Definition 1.3.2 (Riemann Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall (P, \xi)\ \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Remark (The one symbol that changed everything). Cauchy quantifies over choppings. Riemann quantifies over choppings and every possible choice of heights:

$$\forall P \qquad \rightsquigarrow \qquad \forall (P, \xi).$$

That extra quantifier removes the arbitrary left endpoint and turns tag independence into the price of admission.

Remark (Payoff). The definition is no longer tied to continuity. It can admit bounded functions with discontinuities, because the question is not whether f is continuous but whether all fine tagged sums are forced into one narrow numerical corridor. The Cauchy criterion below makes that statement intrinsic: the value need not be known in advance.

Remark (The ruler function example). Thomae's function, also called the ruler function, is

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational, so its discontinuities are dense. Nevertheless its Riemann integral on any compact interval is 0. The reason is that the rational points where T can be large form only a finite set above any fixed height threshold. Fine partitions can isolate those finitely many tall spikes, while all remaining slabs have uniformly small height. This is exactly the kind of function Cauchy's continuity proof could not diagnose, but Riemann's tag-independent definition can admit.

Theorem (Continuous Implies Riemann-Integrable)

Theorem 1.3.3 (Continuous Functions Are Riemann-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem (Linearity)

Theorem 1.3.4 (Linearity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\, dx = \alpha \int_a^b f\, dx + \beta \int_a^b g\, dx.$$

Theorem (Monotonicity)

Theorem 1.3.5 (Monotonicity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\, dx \le \int_a^b g\, dx.$$

Theorem (Triangle Inequality)

Theorem 1.3.6 (Triangle Inequality for the Riemann Integral)**.** *If f is Riemann-integrable on $[a,b]$, then $|f|$ is Riemann-integrable and*

$$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx.$$

Theorem (Interval Additivity)

Theorem 1.3.7 (Interval Additivity for the Riemann Integral)**.** *If $c \in [a,b]$, then f is Riemann-integrable on $[a,b]$ if and only if it is Riemann-integrable on $[a,c]$ and $[c,b]$, and in that case*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Theorem (Cauchy Criterion)

Theorem 1.3.8 (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P,ξ) and (Q,η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f,P,\xi) - S(f,Q,\eta)| < \varepsilon.$$

Remark (Bug report)*.* Riemann fixed the arbitrary tag, but the definition is operationally hostile. To verify integrability directly one must control all tagged partitions at once. The patch request is to remove tags from the verification problem: replace every possible sample height by the slab infimum and supremum, then squeeze.

Remark (Why this is hard to use)*.* Riemann is logically clean but not mechanically friendly. Refining a partition does not make an arbitrary Riemann sum move monotonically, because every refinement lets the tags move again. There is no single upper or lower quantity improving under refinement. Darboux supplies exactly that missing monotone control.

1.4 The Darboux Integral

Geometric Intuition

Darboux stops chasing tags. On each slab, the adversary can do no better than the supremum and no worse than the infimum. So build the upper staircase from $\sup f$, the lower staircase from $\inf f$, and ask whether the gap can be squeezed to zero.

Remark (Engineering reading)*.* Darboux is the usable face of Riemann integration. It does not change the integral; it changes the test. Instead of quantifying over every possible tag, it brackets all tags at once.

Definition (Darboux Sums)

Definition 1.4.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a, b]$, set

$$m_i = \inf_{[x_{i-1},x_i]} f, \qquad M_i = \sup_{[x_{i-1},x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i.$$

Lemma (Monotonicity Under Refinement)

Lemma 1.4.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P).$$

Remark (Geometric reading). Thinner slabs give lower rectangles room to rise and upper rectangles room to fall. This monotonicity is exactly what Riemann sums lack when tags move adversarially.

Remark (One-point refinement proof idea). It is enough to add one cut-point c inside one slab and iterate. The infimum over the whole slab is less than or equal to the infimum over each piece, so the lower contribution rises. The supremum over the whole slab is greater than or equal to the supremum over each piece, so the upper contribution falls.

Definition (Darboux Integrability)

Definition 1.4.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem (Darboux Criterion)

Theorem 1.4.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Remark (The squeeze in one plate). Figure 1.2 places the Cauchy/Riemann tag choices, the lower–upper Darboux bracket, the refinement squeeze, and the gauge panel on the same page. The Darboux row is the chapter's hinge: replacing moving tags by inf and sup turns integrability into the checkable $U - L$ squeeze.

Probes of the Integral

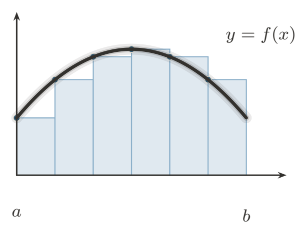

Figure 1. **Left Riemann Sum:**
Tag at the Left Endpoint

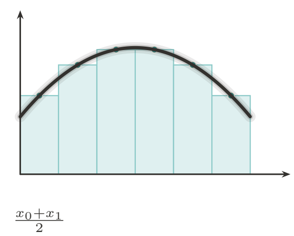

Figure 2. **Midpoint Rule:**
The Centered Tag

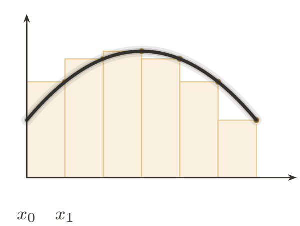

Figure 3. **Right Riemann Sum:**
Tag at the Right Endpoint

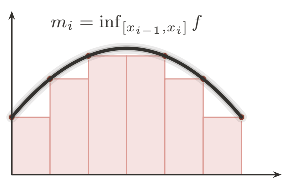

Figure 4. **Lower Darboux Sum:**
The Infimum Probe

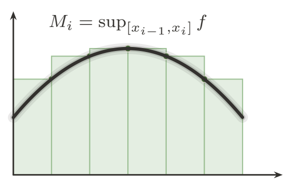

Figure 5. **Upper Darboux Sum:**
The Supremum Probe

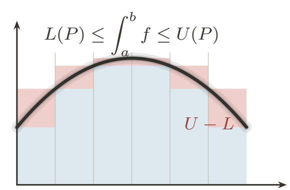

Figure 6. **The Squeeze:**
Integrability as a Closing Gap

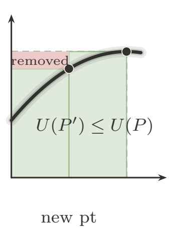

Figure 7. **Partition Refinement:**
Adding Points Tightens the Sums

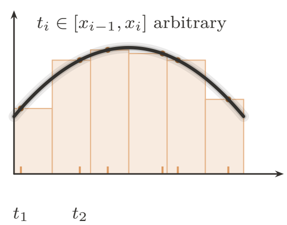

Figure 8. **Tagged Partition:**
The General Riemann Sum

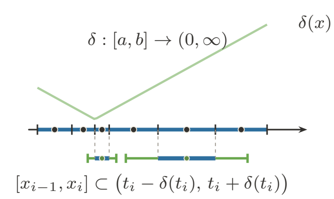

Figure 9. **The Gauge:**
δ-Fine Partitions (Henstock–Kurzweil)

Figure 1.2: Probes of the integral

Theorem (Riemann Equals Darboux)

Theorem 1.4.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a, b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem (Continuous Functions Are Integrable)

Theorem 1.4.6 (Continuous Functions Are Darboux-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem (Monotone Functions Are Integrable)

Theorem 1.4.7 (Monotone Functions Are Darboux-Integrable)**.** *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Remark (Why this is a harvest theorem). Monotone functions may have infinitely many discontinuities. Darboux still handles them because their total vertical movement is finite. On each slab the oscillation is just the endpoint jump across that slab, and these jumps telescope over the partition. Equal-width partitions therefore force $U(f, P) - L(f, P)$ to zero.

Theorem (Finitely Many Discontinuities)

Theorem 1.4.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Remark (Why finite bad sets are harmless). The Darboux gap is

$$U(f, P) - L(f, P) = \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i.$$

A finite set of discontinuities can be trapped in intervals of arbitrarily small total width. On the complement, f is continuous on compact pieces, so its oscillation is uniformly small on fine slabs. Thus the bad slabs are short and the good slabs are flat.

Theorem (Algebra of Darboux-Integrable Functions)

Theorem 1.4.9 (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem (Products)

Theorem 1.4.10 (Products of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem (Absolute Values)

Theorem 1.4.11 (Absolute Values of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem (Continuous Composition)

Theorem 1.4.12 (Continuous Images of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

Remark (Why the harvest is cheap). The Darboux criterion reduces closure questions to controlling the upper-lower gap. Boundedness controls coefficients and products, while continuity of φ converts small oscillation of f into small oscillation of $\varphi \circ f$. This is the point of switching from tags to a squeeze: theorems become estimates on $U(f, P) - L(f, P)$.

Remark (Bug report). Darboux is a better tool for the same class. It does not enlarge Riemann integrability. The remaining defects now fork. One can change the ruler Δx to $\Delta\alpha$, which leads to Riemann–Stieltjes. Or one can diagnose the size of the discontinuity set, which leads to Lebesgue and also to the gauge-integral cure.

Remark (Residual failures). The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ has infimum 0 and supremum 1 on every slab, since rationals and irrationals are dense. Thus $L(f, P) = 0$ and $U(f, P) = 1$ for every partition, so the Darboux gap never closes. This is not a Darboux-specific failure; by equivalence, it is the failure of the whole Riemann family. The failure modes in Figure 1.3 separate the common pathologies: Dirichlet breaks the continuity-almost-everywhere condition, Thomae survives it, unbounded examples violate the boundedness hypothesis, and pointwise limits show why Riemann integration has weak limit behavior.

Where Riemann Fails

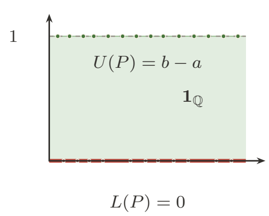

Figure 1. **Dirichlet $\mathbf{1}_{\mathbb{Q}}$: The Gap That Never Closes**

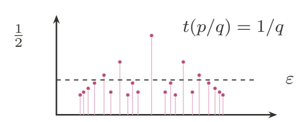

Figure 2. **Thomae's Function: Dense Jumps, Yet Integrable**

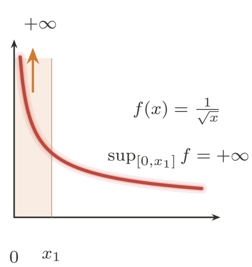

Figure 3. **Unbounded Integrand: Boundedness Is Not Optional**

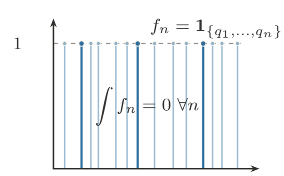

Figure 4. **Pointwise Limit:** $\lim \int \neq \int \lim$ **(Lebesgue's Cue)**

Figure 1.3: Failure modes of the Riemann family

Remark (What Darboux makes visible). For Thomae's function, Darboux sees the opposite picture. On every slab the infimum is 0, because every slab contains irrationals. The supremum can be large only near rationals with small denominator; above any fixed height threshold, there are only finitely many such rationals. A partition can isolate those finitely many tall rational spikes inside intervals of tiny total width, and the remaining upper rectangles are short. Thus the upper sum can be forced below any prescribed ε, while every lower sum is 0.

This is the practical gain: Riemann gives the correct definition, but Darboux gives the hand tool. Prove $U - L$ can be squeezed, and the Riemann value follows from the equivalence theorem.

Remark (The fork)*.* Darboux exposes two separate defects:

Residual defect	Successor	Change
only ordinary length	Riemann–Stieltjes	$\Delta x_i \mapsto \Delta\alpha_i$
domain slabs see every bad set	Lebesgue / HK	measure bad sets or adapt fineness

Stieltjes is therefore a sideways generalization, not the final repair of Riemann's domain-partition limitation.

1.5 The Riemann–Stieltjes Integral

Geometric Intuition

Riemann and Darboux weigh each slab by ordinary length, Δx_i. Riemann–Stieltjes changes the ruler. A slab is weighed by how much an integrator α changes across it:

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

If $\alpha(x) = x$, ordinary Riemann integration returns. If α jumps, the integral sees a point mass.

Remark (Engineering reading)*.* Stieltjes does not repair the discontinuity-set problem. It answers a different complaint: ordinary integration has only one ruler. By replacing Δx_i with $\Delta\alpha_i$, the integral can count regions with nonuniform weight and can encode discrete masses as jumps of α.

Definition (Total Variation)

Definition 1.5.1 (Total Variation)**.** The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition (Bounded Variation)

Definition 1.5.2 (Bounded Variation)**.** The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition (Monotone Implies BV)

Proposition 1.5.3 (Monotone Functions Have Bounded Variation)**.** *If α is monotone on $[a, b]$, then α has bounded variation.*

Remark (Why bounded variation is the honest ruler condition)*.* If α is increasing, the increments telescope:

$$\sum_i \big(\alpha(x_i) - \alpha(x_{i-1})\big) = \alpha(b) - \alpha(a).$$

Bounded variation is the signed analogue of that finite ruler budget. It is what prevents the weighted sums from accumulating uncontrolled oscillatory ruler error.

Definition (Riemann–Stieltjes Integral)

Definition 1.5.4 (Riemann–Stieltjes Integral)**.** The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f \, d\alpha$.

Theorem (Continuous Integrand, BV Integrator)

Theorem 1.5.5 (Continuous Integrand and BV Integrator)**.** *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f \, d\alpha$ exists.*

Theorem (Bilinearity)

Theorem 1.5.6 (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g) \, d\alpha = \lambda \int_a^b f \, d\alpha + \mu \int_a^b g \, d\alpha$$

and

$$\int_a^b f \, d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f \, d\alpha + \mu \int_a^b f \, d\beta.$$

Theorem (Interval Additivity)

Theorem 1.5.7 (Interval Additivity for the Riemann–Stieltjes Integral)**.** *If $c \in [a, b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f \, d\alpha = \int_a^c f \, d\alpha + \int_c^b f \, d\alpha.$$

Theorem (Integration by Parts)

Theorem 1.5.8 (Riemann–Stieltjes Integration by Parts). *If f and α are such that one of $\int_a^b f\,d\alpha$ and $\int_a^b \alpha\,df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem (Smooth Integrator Reduction)

Theorem 1.5.9 (Reduction to Riemann Integration for a C^1 Integrator). *If $\alpha \in C^1([a,b])$ and f is Riemann-integrable on $[a,b]$, then*

$$\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx.$$

Theorem (Step Integrator)

Theorem 1.5.10 (Step Integrators Give Finite Sums). *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

Remark (Proof idea). The Cauchy common-refinement argument runs again. Uniform continuity makes $\omega_f(I)$ small on short slabs, and bounded variation controls $\sum_i |\Delta\alpha_i|$. The error has the form

$$\sum_i \omega_f(I_i)\,|\Delta\alpha_i|,$$

so small oscillation multiplied by finite total variation is small.

Remark (Sums as integrals). If α is a step function with jumps at points c_k, then the Stieltjes integral collapses to a weighted sum of the values $f(c_k)$ times the jumps. This is the conceptual doorway to probability: a cumulative distribution function is an integrator, and expectation is an integral against that integrator.

Remark (Step integrator calculation). Suppose α is constant except for finitely many jumps $\Delta\alpha(c_k) = \alpha(c_k+) - \alpha(c_k-)$. Any sufficiently fine partition has one tiny slab around each c_k carrying that jump, while all other slabs have $\Delta\alpha_i = 0$. If f is continuous at each c_k, then the tagged height on the k-th jump slab is forced toward $f(c_k)$, so

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

This is why a finite weighted sum is not a different species from an integral: it is a Stieltjes integral against a discrete ruler.

Remark (Shared-jump failure). The new pathology is real. On $[0,1]$, let $c = 1/2$, let

$$\alpha(x) = \mathbf{1}_{[c,1]}(x), \qquad f(x) = \mathbf{1}_{[c,1]}(x).$$

The only nonzero $\Delta\alpha_i$ occurs on the slab containing c. If the tag for that slab is chosen left of c, its contribution is 0; if the tag is chosen at or right of c, its contribution is 1. This discrepancy does not shrink with the slab width, because the ruler has a whole unit of mass concentrated at the jump. Tag independence fails, so the Riemann–Stieltjes integral does not exist in this convention.

Remark (Bug report). Stieltjes is a sideways generalization, not the Lebesgue fix. Shared discontinuities of f and α can destroy tag independence, and the standard existence theorem leans on the finite ruler budget $V_a^b(\alpha) < \infty$. Brownian paths have infinite variation almost surely, so stochastic integration cannot be ordinary pathwise Riemann–Stieltjes integration.

Remark (Patch request). The next stochastic patch is not "make Stieltjes a little better." Brownian motion forces a different limiting mode: an L^2 or probabilistic limit rather than a pathwise Riemann–Stieltjes limit. That is the structural reason Ito integration is not just Stieltjes integration with a rougher integrator.

1.6 The Henstock–Kurzweil Integral

Where This Sits

Darboux exposed the Riemann disease before measure theory names it precisely: one uniform fineness parameter must serve the whole interval. There are two ways forward. Lebesgue will change what is measured. The gauge integrals keep the tagged-sum picture and change exactly one symbol: the constant fineness $\delta > 0$ becomes a positive function $\delta(x)$.

Remark (Sibling cure). Henstock–Kurzweil belongs here, before McShane and the final measure-zero diagnosis, because it is still visibly a Riemann-style tagged-sum integral. It patches the uniform-mesh defect without abandoning slabs, tags, or sums. McShane will modify the tag rule one more time, producing a gauge-sum version of Lebesgue integration.

Geometric Intuition

A gauge is adaptive mesh refinement written as analysis. It is tiny where the function needs tight local control and generous where the function is calm. A tagged slab is allowed only if it fits inside the local radius prescribed by its own tag.

Remark (The one-symbol upgrade). Riemann uses a positive number:

$$\exists \delta > 0.$$

Henstock–Kurzweil uses a positive function:

$$\exists \delta : [a,b] \to (0,\infty).$$

The mesh condition $\|P\| < \delta$ becomes the local condition that each slab is δ-fine at its tag.

Definition (Gauge)

Definition 1.6.1 (Gauge on an Interval)**.** A *gauge* on $[a, b]$ is a function

$$\delta : [a, b] \to (0, \infty).$$

Definition (Delta-Fine Tagged Partition)

Definition 1.6.2 (δ-Fine Tagged Partition)**.** A tagged partition (P, ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition (Henstock–Kurzweil Integral)

Definition 1.6.3 (Henstock–Kurzweil Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma (Cousin)

Lemma 1.6.4 (Cousin's Lemma)**.** *For every gauge δ on $[a, b]$, there exists a δ-fine tagged partition of $[a, b]$.*

Remark (Non-vacuity). Cousin's Lemma prevents the definition from being empty. No matter how wildly the gauge varies, compactness of $[a, b]$ guarantees that at least one tagged partition respects all the local step-size demands.

Theorem (Riemann Sits Inside Henstock–Kurzweil)

Theorem 1.6.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable)**.** *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, and the two integrals have the same value.*

Remark (Constant gauges). Riemann is the special case where the gauge is constant. If a Riemann proof hands over one mesh tolerance δ_0, the gauge proof uses $\delta(x) = \delta_0$ for every x.

Lemma (Straddle)

Lemma 1.6.6 (Straddle Lemma)**.** *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - u).$$

Theorem (FTC for the Gauge Integral)

Theorem 1.6.7 (Fundamental Theorem for the Henstock–Kurzweil Integral)**.** *If* $F : [a, b] \to \mathbb{R}$ *is differentiable at every point and* $F' = f$*, then* f *is Henstock–Kurzweil integrable and*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Theorem (Continuous Functions Are Henstock–Kurzweil Integrable)

Theorem 1.6.8 (Continuous Functions Are Henstock–Kurzweil Integrable)**.** *Every continuous function* $f : [a, b] \to \mathbb{R}$ *is Henstock–Kurzweil integrable.*

Remark (What it buys)*.* On a compact interval,

$$\mathcal{R}[a, b] \subsetneq \mathcal{M}[a, b] = \mathcal{L}[a, b] \subsetneq \mathcal{HK}[a, b].$$

Figure 1.4 records this inclusion lattice and the standard witness at each frontier. The first strict inclusion is witnessed by functions like $\mathbf{1}_{\mathbb{Q}}$, which are Lebesgue-integrable but not Riemann integrable. The second is witnessed by derivatives such as that of $F(x) = x^2 \sin(1/x^2)$ with $F(0) = 0$: the derivative is HK-integrable by the gauge FTC even when its absolute value is not Lebesgue-integrable.

This inclusion statement anticipates the next volume's Lebesgue theory. The present chapter has not built Lebesgue integration yet; it has only isolated the Riemann-side machinery, the gauge machinery, and the reason a measure-theoretic successor is needed.

The Integration Lattice

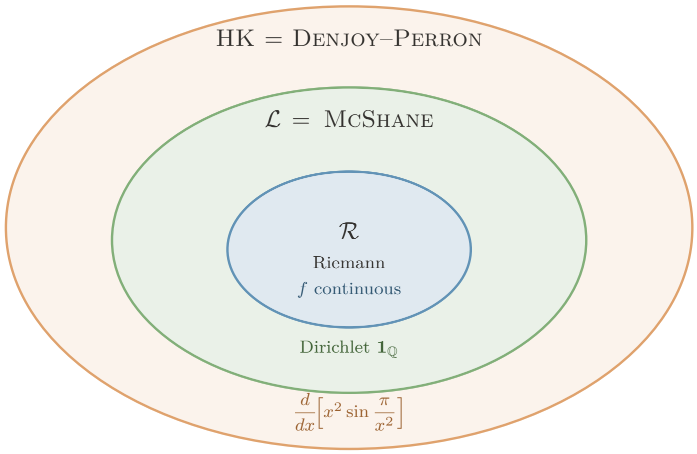

Figure. The Integration Lattice: each frontier is crossed by exactly one witness.

$\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$, with the strict gaps witnessed by:

- $\mathcal{L} \setminus \mathcal{R}$: $\mathbf{1}_{\mathbb{Q}}$ — Lebesgue (and McShane) integrable, $\int = 0$; not Riemann.
- $\mathcal{HK} \setminus \mathcal{L}$: F' for $F(x) = x^2 \sin(\pi/x^2)$, $F(0) = 0$ — HK-integrable ($\int_0^1 F' = F(1) - F(0) = 0$) but $\int_0^1 |F'| = \infty$, so not Lebesgue.

Engine: $\mathcal{L}$ = McShane uses *free* tags (absolute integration); $\mathcal{HK}$ uses *tethered* tags, which preserve cancellation — the source of the outermost ring, and the reason HK integrates *every* everywhere-derivative.

Figure 1.4: The inclusion lattice for Riemann, Lebesgue–McShane, and Henstock–Kurzweil integration

Remark (Killer example). Let $F(x) = x^2 \sin(1/x^2)$ for $x \neq 0$, and set $F(0) = 0$. Then F is differentiable at 0, with $F'(0) = 0$, while for $x \neq 0$,

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2).$$

The oscillatory term has nonintegrable absolute size near 0, so this derivative is the standard witness that HK reaches beyond Lebesgue while the FTC remains exact.

Remark (HK's own bug report). Henstock–Kurzweil is not simply "the best integral." Lebesgue has stronger limit theorems, a complete L^1 normed-space home, and a clean abstract measure-space setting. HK is the jewel for the one-dimensional FTC and conditional accumulation; Lebesgue and Ito remain the engines for probability and stochastic integration.

Remark (Computational bridge). A gauge $\delta(x)$ is a spatially varying step-size field. Lipschitz control is the case where a constant gauge is enough; singular or non-Lipschitz regions demand a genuinely varying gauge. Cousin's Lemma is the compactness guarantee that a valid adaptive partition exists.

Remark (One-page summary).

Integral	Patch	Remaining defect
Cauchy	left rectangles settle for continuous f	privileged tag
Riemann	all tags must agree	hard to verify
Darboux	sup/inf squeeze	same class as Riemann
Stieltjes	change the ruler	needs BV; shared jumps
Henstock–Kurzweil	local gauge fineness	weaker limit machinery
McShane	free gauge tags	same class as Lebesgue
Lebesgue	measure level sets	abstract machinery required

The thread is the right-hand column: every integral is a direct response to a specific defect.

1.7 The McShane Integral

Where This Sits

McShane is the gauge integral that looks almost indistinguishable from Henstock–Kurzweil until one notices where the tags are allowed to live. HK keeps each tag inside its own slab. McShane lets a tag control a slab from outside it, as long as the slab lies inside the tag's gauge ball.

Remark (The tiny syntactic change). Henstock–Kurzweil says:

$$\xi_i \in [x_{i-1}, x_i] \quad \text{and} \quad [x_{i-1}, x_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)).$$

McShane drops only the first requirement. The interval still has to be locally small around its tag, but the tag need not lie in the interval it controls. That small freedom is decisive: it prevents the conditional cancellation that lets HK integrate every derivative.

Remark (HK versus McShane at a glance).

Integral	Gauge	Tag position	Class on $[a, b]$
Henstock–Kurzweil	$\delta(x)$	$\xi_i \in I_i$	$\mathcal{HK}$
McShane	$\delta(x)$	$\xi_i \in [a, b]$ and $I_i \subset B(\xi_i, \delta(\xi_i))$	$\mathcal{L}$

Both use a variable gauge. The difference is not the step-size field; it is whether the tag must sit inside the interval whose height it supplies. Figure 1.5 shows this tethered-versus-free distinction directly.

Two Ways to Tag

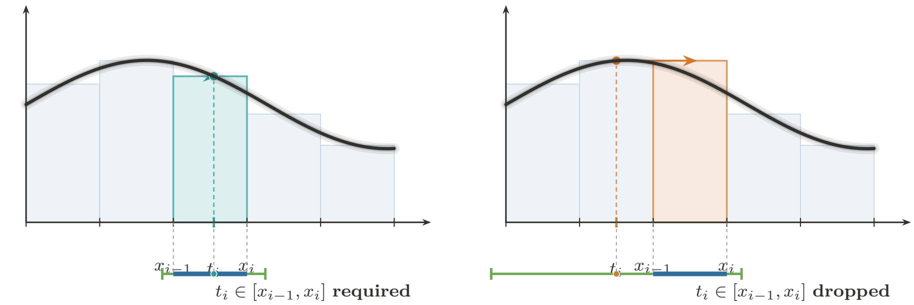

Figure 1. **Henstock–Kurzweil: Tag Tethered to Its Cell**

Figure 2. **McShane: The Tag Roams Free**

Same integrand, same gauge condition $I_i \subseteq (t_i - \delta(t_i),\, t_i + \delta(t_i))$. The *only* difference is whether the tag must sit in its own cell.

HK keeps the tether $\Rightarrow$ conditional convergence survives $\Rightarrow$ $\mathcal{HK}$ = Denjoy–Perron (nonabsolute).
McShane cuts it $\Rightarrow$ only absolute integrals remain $\Rightarrow$ McShane = $\mathcal{L}^1$ (Lebesgue).

Figure 1.5: Henstock–Kurzweil versus McShane fineness

Geometric Intuition

In HK, a tag is a sample point inside the slab. In McShane, a tag is more like a local address that authorizes nearby slabs. The gauge still says how far the tag's influence reaches, but the rectangle height may be chosen from a point near the slab rather than inside it. This makes the integral less forgiving about sign-changing singular behavior and much closer to Lebesgue's absolute notion of size.

Definition (McShane-Fine Tagged Partition)

Definition 1.7.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a, b] \to (0, \infty)$ be a gauge. A finite tagged partition $\{([u_i, v_i], \xi_i)\}_{i=1}^n$ is *McShane δ-fine* if the intervals $[u_i, v_i]$ are non-overlapping, cover $[a, b]$, each $\xi_i \in [a, b]$, and

$$[u_i, v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i, v_i]$.

Remark (Untethered tag). Figure 1.6 isolates the McShane move: a tag may sit outside the cell whose height it supplies, provided the cell lies inside the tag's gauge ball. The ruler is still Δx_i; the change is entirely in the admissible tag geometry.

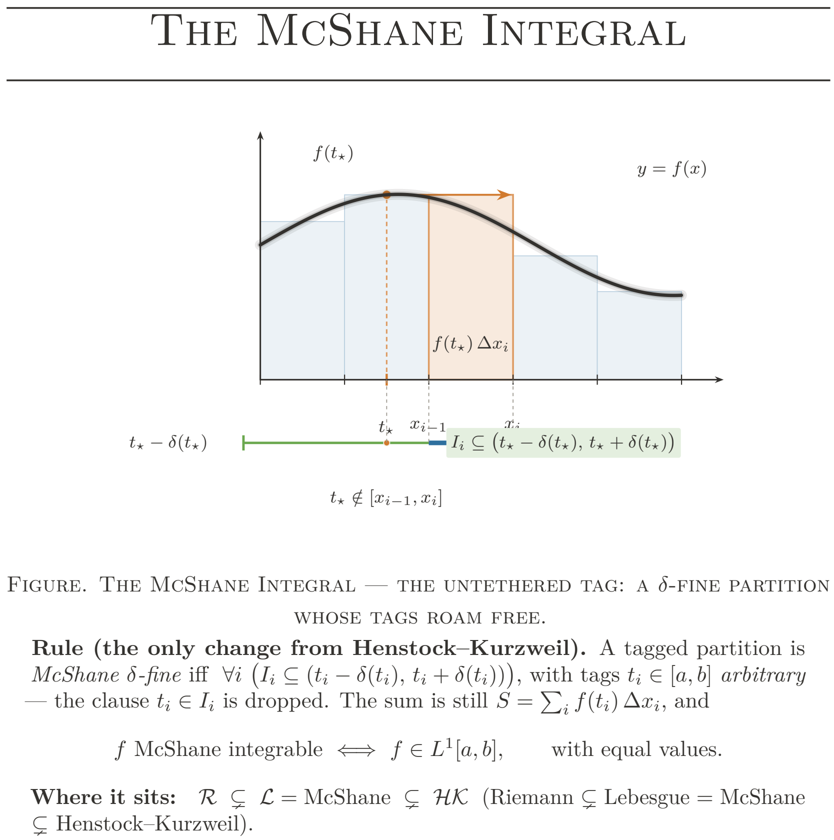

Figure 1.6: The McShane integral and the untethered tag

Definition (McShane Integral)

Definition 1.7.2 (McShane Integral)**.** A function $f : [a,b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a,b] \to (0,\infty)\ \forall (P,\xi)\ \big((P,\xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f,P,\xi) - A| < \varepsilon\big),$$

where

$$S(f,P,\xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Remark (Relation to HK)*.* Every McShane-fine partition is not necessarily HK-fine, because its tags may float outside their intervals. But every HK-fine partition is McShane-fine. Therefore McShane integrability is a stronger demand: the sums must survive all the usual gauge-fine tests and also the extra free-tag tests.

Theorem (McShane Sits Between Riemann and HK)

Theorem 1.7.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a,b] \subseteq \mathcal{M}[a,b] \subseteq \mathcal{HK}[a,b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem (McShane Equals Lebesgue)

Theorem 1.7.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b],$$

and the McShane and Lebesgue integral values agree.

Remark (The Lebesgue connection)*.* For real-valued functions on a compact interval, the McShane integral recovers the Lebesgue integral. In practical terms,

$$\mathcal{M}[a,b] = \mathcal{L}[a,b]$$

with equal values. Thus McShane is not a second route beyond Lebesgue in the way HK is. It is a gauge-sum presentation of Lebesgue integration.

Remark (Comparison chain)*.* Combining the Riemann inclusion, the McShane–Lebesgue identification, and the HK derivative witness gives

$$\mathcal{R}[a,b] \subsetneq \mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b] \subsetneq \mathcal{HK}[a,b].$$

The first strict inclusion is the old Riemann failure at dense null discontinuities; the second is the HK recovery of derivatives whose absolute size is not Lebesgue integrable.

Remark (Forward dependency)*.* The proof that every Riemann-integrable function is McShane-integrable uses the measure-zero diagnosis in the next section: Riemann integrability means the discontinuity set is null. The exposition places McShane first because it belongs syntactically with the gauge integrals; the proof ledger records that it imports the Lebesgue criterion.

Remark (Why free tags force Lebesgue behavior)*.* The HK derivative example

$$F'(x) = 2x\sin(1/x^2) - \frac{2}{x}\cos(1/x^2)$$

is conditionally integrable by HK because the tags remain inside the tiny slabs where the derivative is being locally linearized. McShane's free tags can sample nearby oscillatory peaks in a way that destroys this conditional balance. What survives that stronger test is exactly the Lebesgue-integrable, absolutely controlled part of the gauge world.

Remark (What it buys)*.* McShane gives a tagged-sum construction of the Lebesgue integral without starting from simple functions or sigma-algebras. Conceptually, that is its job in this chapter: it shows that adaptive Riemann-style sums can reach the Lebesgue class, while HK shows that keeping tags inside slabs reaches beyond Lebesgue to conditionally integrable derivatives.

Remark (Bug report). McShane is elegant but not the terminal fix for this chapter's story. If one wants a tagged-sum definition of the Lebesgue integral, it is excellent. If one wants the perfect one-dimensional Fundamental Theorem of Calculus, HK is strictly stronger. And if one wants probability, stochastic processes, and abstract spaces, the next volume still needs measure theory explicitly.

Remark (Handoff to measure zero). McShane reveals that gauge machinery and Lebesgue size are compatible. The next section names the exact size condition already controlling the older Riemann–Darboux family: the discontinuity set must have measure zero.

1.8 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Darboux integrability fails where upper and lower staircases refuse to meet. That refusal is local oscillation. Thus the exact Riemann question becomes: how large is the set where f is discontinuous?

Remark (Engineering reading). This final section is not another integral. It is the diagnostic scan of the entire Riemann–Darboux family, placed after Henstock–Kurzweil and McShane so the chapter ends by naming the exact obstruction that opens the Lebesgue volume. Once Darboux has made the gap visible, the remaining question is whether the bad part of the domain is small enough to cover by slabs of negligible total length.

Definition (Measure Zero)

Definition 1.8.1 (Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition (Point Oscillation)

Definition 1.8.2 (Point Oscillation)**.** For bounded f on $[a, b]$, define

$$\omega_f(x) = \inf_{\delta>0} \omega_f\big((x-\delta, x+\delta) \cap [a, b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Remark (Oscillation levels). The sets

$$D_\eta = \{x : \omega_f(x) \geq \eta\}$$

separate the bad set by severity. The full discontinuity set is

$$D = \bigcup_{k=1}^{\infty} D_{1/k}.$$

The proof of the criterion works by showing that positive-size high oscillation forces a permanent Darboux gap, while null high-oscillation sets can be covered by intervals of tiny total length.

Theorem (Lebesgue Criterion)

Theorem 1.8.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

has measure zero.

Remark (Terminal diagnosis)*.* Riemann integration works precisely when the set of bad points is invisible to length. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ is discontinuous everywhere, so $D = [0, 1]$, not a null set. The Lebesgue patch is to stop partitioning the domain into slabs and instead measure the sets where the function takes given ranges of values. Henstock–Kurzweil keeps tagged sums, but replaces uniform fineness by local fineness; McShane shows that a slightly stricter gauge rule recovers the Lebesgue class itself. The pathology plate in Figure 1.3 is the visual companion to this diagnosis.

Remark (Lebesgue handoff)*.* The slogan is precise: Riemann partitions the domain, so every slab that meets a dense bad set still sees the bad behavior. Lebesgue partitions the range: it groups points by height and measures the set carrying each height. For $\mathbf{1}_{\mathbb{Q}}$, the height 1 lives on a null set and therefore contributes nothing.

Appendix: Pathologies of the Riemann Integral

Remark (Source placement)*.* This appendix is the converted support note from `integration-failures.tex`. Its standalone theorem machinery and document preamble have been removed; the material is kept as commentary so it does not duplicate the formal Lebesgue criterion in Section 1.8.3.

Remark (The one theorem behind the failures)*.* The Riemann failures in Figure 1.3 are not a bag of examples. They are tests against the two hypotheses in the Lebesgue criterion: boundedness, and a discontinuity set of measure zero. Dirichlet fails because its discontinuity set is all of $[0, 1]$. Unbounded examples fail before the criterion even begins. Pointwise limits expose a different weakness: the Riemann class is not stable under the limit operations that Lebesgue integration was built to support.

Remark (Thomae as the diagnostic example)*.* Thomae's function is the useful warning. Its discontinuities are dense, but they are countable and therefore null. Darboux can isolate the finitely many large spikes above any fixed threshold inside intervals of tiny total length;

outside those intervals, the upper sums are uniformly small. Thus density of the bad set is not the obstruction. Size is.

Remark (Through-line). The atlas failure note should be read as the bridge from Darboux to Measure Zero: $U(f, P) - L(f, P)$ is local oscillation times length, so the only bad oscillation that matters is bad oscillation carried by a set of positive length. That observation closes the Riemann family and points directly to Lebesgue's range-partition construction.

Appendix: The Gauge Integrals

Remark (Source placement). This appendix is the converted support note from `gauge-integrals.tex`. The standalone preamble, local theorem declarations, and duplicate figure includes have been removed; the shared figures are the pinned figures already included in Sections 1.6 and 1.7.

Remark (The gauge framework). The gauge integrals replace Riemann's single global mesh tolerance by a local step-size controller $\delta : [a, b] \to (0, \infty)$. Henstock–Kurzweil and McShane then differ by one geometric clause. HK requires the tag to sit inside the interval it controls. McShane drops that tether and allows the tag to range over $[a, b]$, provided the interval lies inside the tag's gauge ball. The contrast is visualized in Figures 1.5 and 1.6.

Remark (Free tags and Lebesgue). McShane's free tags force absolute, Lebesgue-style behavior. Since a tag can authorize nearby intervals without lying inside them, the sums can test whether the mass assigned to height bands is genuinely controlled. On compact intervals this recovers the Lebesgue integral: $\mathcal{M}[a, b] = \mathcal{L}[a, b]$, with equal values.

Remark (Tethered tags and the FTC). HK keeps the tag tethered to its interval, which preserves local cancellation for derivatives. That is why the HK integral has the exact one-dimensional FTC: if F is differentiable everywhere, then F' is HK-integrable and (HK) $\int_a^b F' = F(b) - F(a)$, even when F' is not Lebesgue integrable.

Remark (Lattice). The resulting compact-interval lattice is

$$\mathcal{R}[a, b] \subsetneq \mathcal{L}[a, b] = \mathcal{M}[a, b] \subsetneq \mathcal{HK}[a, b],$$

as recorded in Figure 1.4. This appendix is kept after the main chapter because the book has not yet built the measure theory needed to prove the McShane–Lebesgue identification.

Proofs

Remark (Return). Return to Lemma

Lemma (Common Refinement of Two Partitions). *If P and P' are partitions of $[a, b]$, then $P \cup P'$, ordered in increasing order, is a partition of $[a, b]$ that refines both.*

Professional Standard Proof.
Both partitions are finite subsets of $[a, b]$ containing a and b. Their union is finite, contains a and b, and becomes a strictly increasing list after duplicate points are removed. Since $P \subseteq P \cup P'$ and $P' \subseteq P \cup P'$, the ordered union refines both partitions. □

Detailed Learning Proof.
The only possible obstruction is ordering. A finite set of real numbers can be listed increasingly, and duplicate common nodes are written once. The first node is still a, the last node is still b, and every old node from both partitions remains present. □

Remark (Proof structure). Order the finite union and observe that both original partitions are subsets of it.

Remark (Dependencies).

- Partition of a Compact Interval.

Remark (Return). Return to Proposition

Proposition (Cauchy Integral of a Constant)**.** *If* $f(x) = c$ *on* $[a, b]$*, then* $\int_a^b c\,dx = c(b - a)$*.*

Professional Standard Proof.
For every partition P, $C(f, P) = \sum_i c\Delta x_i = c(b - a)$. The net is constant, so its limit is $c(b - a)$. ☐

Detailed Learning Proof.
The rectangle heights are all c, and the subinterval widths telescope to the total length $b - a$. ☐

Remark (Proof structure). Compute every finite sum and observe it is independent of the partition.

Remark (Dependencies).

- Left-Endpoint Cauchy Sum.

Remark (Return). Return to Theorem

Theorem (Linearity of the Cauchy Integral)**.** *If f and g are Cauchy-integrable, then the integral is linear.*

Professional Standard Proof.
For each partition, $C(\alpha f + \beta g, P) = \alpha C(f, P) + \beta C(g, P)$. Pass to the existing limits. □

Detailed Learning Proof.
Linearity is already true for finite sums. The integral is the limit of those finite sums, so the same algebra survives the limiting process. □

Remark (Proof structure). Prove linearity at the sum level, then pass to limits.

Remark (Dependencies).

- Left-Endpoint Cauchy Sum.
- Cauchy Integral.

Remark (Return). Return to Theorem

Theorem (Monotonicity of the Cauchy Integral). *If $f \leq g$ pointwise and both functions are Cauchy-integrable, then the integral of f is no larger than the integral of g.*

Professional Standard Proof.
TODO: professional standard proof for monotonicity of the Cauchy integral. □

Detailed Learning Proof.
TODO: detailed learning proof for monotonicity of the Cauchy integral. □

Remark (Proof structure).

Remark (Dependencies).

- Cauchy Integral

Remark (Return). Return to Corollary

Corollary (Bounds for the Cauchy Integral). *A continuous function whose values lie between m and M has integral between $m(b-a)$ and $M(b-a)$.*

Professional Standard Proof.
TODO: professional standard proof for bounds for the Cauchy integral. □

Detailed Learning Proof.
TODO: detailed learning proof for bounds for the Cauchy integral. □

Remark (Proof structure).

Remark (Dependencies).

- Cauchy Integral of a Constant
- Monotonicity of the Cauchy Integral

Remark (Return). Return to Theorem

Theorem (Triangle Inequality for the Cauchy Integral)**.** *If f and $|f|$ are Cauchy-integrable, the absolute value of the integral is bounded by the integral of the absolute value.*

Professional Standard Proof.
TODO: professional standard proof for the Cauchy integral triangle inequality. □

Detailed Learning Proof.
TODO: detailed learning proof for the Cauchy integral triangle inequality. □

Remark (Proof structure)*.*

Remark (Dependencies)*.*

- Monotonicity of the Cauchy Integral

Remark (Return). Return to Theorem

Theorem (Interval Additivity for the Cauchy Integral)**.** *When the relevant Cauchy integrals exist, the integral over* $[a, b]$ *splits across an intermediate point* c*.*

Professional Standard Proof.
TODO: professional standard proof for Cauchy integral interval additivity. □

Detailed Learning Proof.
TODO: detailed learning proof for Cauchy integral interval additivity. □

Remark (Proof structure).

Remark (Dependencies).

- Cauchy Integral

Remark (Return). Return to Theorem

Theorem (Continuous Functions Are Cauchy-Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Cauchy-integrable.*

Professional Standard Proof.
Let $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ such that every interval $I \subset [a, b]$ of length less than δ satisfies

$$\omega_f(I) < \frac{\varepsilon}{2(b-a)}.$$

If P and P' have mesh less than δ, let $Q = P \cup P'$ be their common refinement. On each slab $I = [x_{i-1}, x_i]$ of P, refinement only replaces the left-endpoint height $f(x_{i-1})$ by left-endpoint heights inside the same slab. Hence the total change over that slab is bounded by $\omega_f(I)\Delta x_i$. Summing gives

$$|C(f, P) - C(f, Q)| \leq \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i < \frac{\varepsilon}{2}.$$

The same estimate gives $|C(f, P') - C(f, Q)| < \varepsilon/2$, so

$$|C(f, P) - C(f, P')| < \varepsilon.$$

Thus the Cauchy sums form a Cauchy net in $\mathbb{R}$, and completeness gives the limit. □

Detailed Learning Proof.
The refinement Q is the comparison device. Directly comparing P and P' is awkward because their slabs do not line up. But both old sums can be compared to the single refined sum over Q. Uniform continuity makes every old slab nearly constant, so replacing one rectangle by several rectangles inside the same slab changes area by at most "oscillation times width." The two errors meet at Q, and the triangle inequality closes the estimate. □

Remark (Proof structure). Use uniform continuity for local oscillation control, compare through a common refinement, then invoke completeness of $\mathbb{R}$.

Remark (Dependencies).

- Common Refinement of Two Partitions.
- Left-Endpoint Cauchy Sum.
- Oscillation on an Interval.

Remark (Return). Return to Theorem

Theorem (Tag Independence for Continuous Functions)**.** *If f is continuous, all tagged sums have the same limit as the left sums.*

Professional Standard Proof.
Let $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ such that $\omega_f(I) < \varepsilon/(b-a)$ for every subinterval I of length less than δ. If $\|P\| < \delta$, then for any tagging ξ,

$$|S(f, P, \xi) - C(f, P)| \leq \sum_i |f(\xi_i) - f(x_{i-1})|\Delta x_i \leq \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i < \varepsilon.$$

Thus every tagged sum has the same limit as the left-endpoint sums. ☐

Detailed Learning Proof.
Inside a short slab, the tag value and the left-endpoint value are close because both points lie in the same small interval. Multiplying by the slab width converts this height error into an area error. Summing over the partition gives at most small oscillation times total length $b - a$. ☐

Remark (Proof structure). Bound the difference between a tagged sum and a left-endpoint sum by the total oscillation over the partition.

Remark (Dependencies).

- Tagged Partition.
- Left-Endpoint Cauchy Sum.
- Oscillation on an Interval.

Remark (Return). Return to Theorem

Theorem (Continuous Functions Are Riemann-Integrable)**.** *Every continuous function on a compact interval is Riemann-integrable.*

Professional Standard Proof.
TODO: professional standard proof for continuous functions being Riemann-integrable. ☐

Detailed Learning Proof.
TODO: detailed learning proof for continuous functions being Riemann-integrable. ☐

Remark (Proof structure).

Remark (Dependencies).

- Tag Independence for Continuous Functions
- Riemann Integral

Remark (Return). Return to Theorem

Theorem (Linearity of the Riemann Integral). *The Riemann integral is linear on its domain of integrable functions.*

Professional Standard Proof.
TODO: professional standard proof for linearity of the Riemann integral. ☐

Detailed Learning Proof.
TODO: detailed learning proof for linearity of the Riemann integral. ☐

Remark (Proof structure).

Remark (Dependencies).

- Riemann Integral

Remark (Return). Return to Theorem

Theorem (Monotonicity of the Riemann Integral). *If $f \leq g$ pointwise and both functions are Riemann-integrable, then the integral of f is no larger than the integral of g.*

Professional Standard Proof.
TODO: professional standard proof for monotonicity of the Riemann integral. □

Detailed Learning Proof.
TODO: detailed learning proof for monotonicity of the Riemann integral. □

Remark (Proof structure).

Remark (Dependencies).

- Riemann Integral

Remark (Return). Return to Theorem

Theorem (Triangle Inequality for the Riemann Integral)**.** *The Riemann integral satisfies the usual absolute-value bound.*

Professional Standard Proof.
TODO: professional standard proof for the Riemann integral triangle inequality. □

Detailed Learning Proof.
TODO: detailed learning proof for the Riemann integral triangle inequality. □

Remark (Proof structure).

Remark (Dependencies).

- Monotonicity of the Riemann Integral

Remark (Return). Return to Theorem

Theorem (Interval Additivity for the Riemann Integral). *Riemann integrability and the integral value split across an intermediate point.*

Professional Standard Proof.
TODO: professional standard proof for Riemann integral interval additivity. □

Detailed Learning Proof.
TODO: detailed learning proof for Riemann integral interval additivity. □

Remark (Proof structure).

Remark (Dependencies).

- Riemann Integral

Remark (Return). Return to Theorem

Theorem (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if sufficiently fine tagged sums are mutually close.*

Professional Standard Proof.
Suppose f is Riemann-integrable with value L. Given $\varepsilon > 0$, choose $\delta > 0$ so that every tagged partition of mesh less than δ has sum within $\varepsilon/2$ of L. Then any two such sums differ by less than ε by the triangle inequality.

Conversely, assume the mutual closeness condition. The tagged sums, directed by refinement and decreasing mesh, form a Cauchy net in $\mathbb{R}$. Since $\mathbb{R}$ is complete, this net converges to some L. The Cauchy condition then says exactly that all sufficiently fine tagged sums are close to this L, which is the Riemann integral. □

Detailed Learning Proof.
If all fine sums are near L, then any two are near each other because both sit in the same small interval around L. The converse uses the same idea as the Cauchy criterion for sequences: a family whose sufficiently late values are mutually close has a limit in a complete space. □

Remark (Proof structure). Use the triangle inequality for the easy direction and completeness for the converse.

Remark (Dependencies).

- Riemann Sum.
- Riemann Integral.

Remark (Return). Return to Lemma

Lemma (Darboux Sums Under Refinement)**.** *Refinement raises lower sums and lowers upper sums.*

Professional Standard Proof.
It suffices to add one point $c \in (x_{k-1}, x_k)$. Let

$$I = [x_{k-1}, x_k], \qquad I_1 = [x_{k-1}, c], \qquad I_2 = [c, x_k].$$

Write m, m_1, m_2 for the infima of f on I, I_1, I_2, and M, M_1, M_2 for the corresponding suprema. Since $I_1, I_2 \subset I$,

$$m \leq m_1, \quad m \leq m_2, \qquad M_1 \leq M, \quad M_2 \leq M.$$

Therefore

$$m(c - x_{k-1}) + m(x_k - c) \leq m_1(c - x_{k-1}) + m_2(x_k - c),$$

so the lower sum rises. Similarly,

$$M_1(c - x_{k-1}) + M_2(x_k - c) \leq M(c - x_{k-1}) + M(x_k - c),$$

so the upper sum falls. Repeating for each added point proves the chain. □

Detailed Learning Proof.
When a slab is split, the new lower rectangles may use larger infima because each smaller interval has fewer points where the function can dip. The new upper rectangles may use smaller suprema for the same reason. This is the monotone control Darboux extracts from refinement. □

Remark (Proof structure). Check the effect of adding one partition point, then iterate.

Remark (Dependencies).

- Lower and Upper Darboux Sums.
- Refinement of a Partition.

Remark (Return). Return to Theorem

Theorem (Darboux Criterion). *Darboux integrability is equivalent to making $U(f, P) - L(f, P)$ arbitrarily small.*

Professional Standard Proof.
Let

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

If $\underline{I} = \overline{I} = I$, choose partitions P_1, P_2 such that

$$I - L(f, P_1) < \varepsilon/2, \qquad U(f, P_2) - I < \varepsilon/2.$$

Let $Q = P_1 \cup P_2$. By refinement monotonicity,

$$L(f, Q) \geq L(f, P_1), \qquad U(f, Q) \leq U(f, P_2),$$

so

$$U(f, Q) - L(f, Q) < \varepsilon.$$

Conversely, if for every $\varepsilon > 0$ some P satisfies $U(f, P) - L(f, P) < \varepsilon$, then

$$0 \leq \overline{I} - \underline{I} \leq U(f, P) - L(f, P) < \varepsilon.$$

Since this holds for every $\varepsilon > 0$, $\overline{I} = \underline{I}$. □

Detailed Learning Proof.
The common refinement is what lets the two approximations be used at the same time. One partition pushes the lower sum up near the lower integral; another pushes the upper sum down near the upper integral. Refining them together keeps both improvements. □

Remark (Proof structure). Translate equality of upper and lower integrals into partitions with small upper-minus-lower gap, and conversely.

Remark (Dependencies).

- Lower and Upper Darboux Sums.
- Darboux Integrability.

Remark (Return). Return to Theorem

Theorem (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions, Riemann and Darboux integrability are equivalent.*

Professional Standard Proof.
For every tagged partition,

$$L(f, P) \leq S(f, P, \xi) \leq U(f, P).$$

If the Darboux criterion gives a partition P with $U(f, P) - L(f, P) < \varepsilon$, then every tagged sum on P is trapped in an interval of length $< \varepsilon$. Refinements preserve the squeeze, so the Riemann sums have a common limit equal to the Darboux value.

Conversely, suppose f is Riemann-integrable with value L. Choose a mesh tolerance so all fine tagged sums are within ε of L. For a fine partition P, choose tags whose values are within a small error of the suprema M_i, and choose another set of tags whose values are within a small error of the infima m_i. The resulting tagged sums approximate $U(f, P)$ and $L(f, P)$, yet both are close to L. Hence $U(f, P) - L(f, P)$ can be made arbitrarily small, so Darboux integrability follows. □

Detailed Learning Proof.
Darboux brackets every Riemann choice at once. If the bracket is narrow, no tagged sum can escape it. If all tagged sums are forced near one value, then tags chosen near the top and bottom of each slab force the Darboux bracket itself to become narrow. □

Remark (Proof structure)*.* Trap every tagged sum between the lower and upper sums, then compare the collapse of these brackets with tag-independent convergence.

Remark (Dependencies).

- Riemann Integral.
- Darboux Integrability.
- Darboux Criterion.

Remark (Return). Return to Theorem

Theorem (Continuous Functions Are Darboux-Integrable). *Every continuous function on a compact interval is Darboux-integrable.*

Professional Standard Proof.
TODO: professional standard proof for continuous functions being Darboux-integrable. □

Detailed Learning Proof.
TODO: detailed learning proof for continuous functions being Darboux-integrable. □

Remark (Proof structure).

Remark (Dependencies).

- Darboux Criterion
- Heine–Cantor uniform continuity theorem.

Remark (Return). Return to Theorem

Theorem (Monotone Functions Are Darboux-Integrable). *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Professional Standard Proof.
Assume first that f is increasing. Let $\varepsilon > 0$. If $f(b) = f(a)$, then f is constant and the result is immediate. Otherwise, choose a partition $P = \{a = x_0 < \cdots < x_n = b\}$ with mesh

$$\|P\| < \frac{\varepsilon}{f(b) - f(a)}.$$

For an increasing function on $[x_{i-1}, x_i]$,

$$m_i = f(x_{i-1}), \qquad M_i = f(x_i),$$

so

$$U(f, P) - L(f, P) = \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) \Delta x_i.$$

Since every $\Delta x_i \leq \|P\|$,

$$U(f, P) - L(f, P) \leq \|P\| \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \|P\| \left(f(b) - f(a)\right) < \varepsilon.$$

By the Darboux criterion, f is integrable.

If f is decreasing, apply the increasing case to $-f$. Since integrability is preserved by scalar multiplication, f is Darboux-integrable. □

Detailed Learning Proof.
The key point is that a monotone function cannot oscillate back and forth. On each small interval the largest possible vertical error is the change from the left endpoint to the right endpoint. When those vertical changes are summed over the partition, all intermediate terms cancel: the total vertical movement is only $f(b) - f(a)$. Multiplying by the largest slab width gives a bound for the whole Darboux gap. Refining the mesh makes that product as small as desired. □

Remark (Proof structure). For increasing functions, bound the Darboux gap by mesh times total vertical variation. Reduce decreasing functions by negation.

Remark (Dependencies).

- Lower and Upper Darboux Sums.
- Darboux Criterion.

Remark (Return). Return to Theorem

Theorem (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a,b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Professional Standard Proof.
Let $|f| \le M$, and let the discontinuities be $c_1, \dots, c_N$. Fix $\varepsilon > 0$. Choose open intervals $G_1, \dots, G_N$, with $c_j \in G_j$, whose total length is less than

$$\frac{\varepsilon}{4M+1}.$$

Let $G = \bigcup_j G_j$, and let

$$K = [a,b] \setminus G.$$

The function f is continuous at every point of K. Since K is compact, there is $\delta > 0$ such that every subinterval of length less than δ that meets K has oscillation less than

$$\frac{\varepsilon}{2(b-a)}.$$

Choose a partition P with mesh less than δ and containing all endpoints of the intervals G_j.

Split the slabs of P into those contained in G and those meeting K. On slabs contained in G, the oscillation is at most $2M$, and the total width is less than $\varepsilon/(4M+1)$, so their contribution to $U(f,P) - L(f,P)$ is less than $\varepsilon/2$. On slabs meeting K, the oscillation is less than $\varepsilon/(2(b-a))$, and their total width is at most $b-a$, so their contribution is less than $\varepsilon/2$. Hence

$$U(f,P) - L(f,P) < \varepsilon.$$

By the Darboux criterion, f is Darboux-integrable. □

Detailed Learning Proof.
The proof separates the interval into bad neighborhoods and good regions. The bad neighborhoods contain the finitely many discontinuities, but they are chosen with tiny total length. Even if f oscillates as badly as possible there, boundedness keeps the total area error small. Outside those neighborhoods, every point is a continuity point, and compactness turns pointwise continuity into one uniform oscillation scale. A fine partition then makes all good slabs nearly flat. □

Remark (Proof structure). Cover finitely many discontinuities by short intervals, use compactness for uniform oscillation control on the complement, then apply the Darboux criterion.

Remark (Dependencies).

- Oscillation on an Interval.
- Lower and Upper Darboux Sums.
- Darboux Criterion.

Remark (Return). Return to Theorem

Theorem (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *Linear combinations of Darboux-integrable functions are Darboux-integrable.*

Professional Standard Proof.
TODO: professional standard proof for Darboux-integrable linear combinations. □

Detailed Learning Proof.
TODO: detailed learning proof for Darboux-integrable linear combinations. □

Remark (Proof structure).

Remark (Dependencies).

- Darboux Criterion

Remark (Return). Return to Theorem

Theorem (Products of Darboux-Integrable Functions)**.** *Products of Darboux-integrable functions are Darboux-integrable.*

Professional Standard Proof.
TODO: professional standard proof for products of Darboux-integrable functions. □

Detailed Learning Proof.
TODO: detailed learning proof for products of Darboux-integrable functions. □

Remark (Proof structure).

Remark (Dependencies).

- Darboux Criterion

Remark (Return). Return to Theorem

Theorem (Absolute Values of Darboux-Integrable Functions)**.** *The absolute value of a Darboux-integrable function is Darboux-integrable.*

Professional Standard Proof.
TODO: professional standard proof for absolute values of Darboux-integrable functions. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute values of Darboux-integrable functions. □

Remark (Proof structure).

Remark (Dependencies).

- Darboux Criterion

Remark (Return). Return to Theorem

Theorem (Continuous Images of Darboux-Integrable Functions)**.** *Continuous functions preserve Darboux integrability under composition with a bounded Darboux-integrable function.*

Professional Standard Proof.
TODO: professional standard proof for continuous composition preserving Darboux integrability. □

Detailed Learning Proof.
TODO: detailed learning proof for continuous composition preserving Darboux integrability. □

Remark (Proof structure).

Remark (Dependencies).

- Darboux Criterion

Remark (Return). Return to Proposition

Proposition (Monotone Functions Have Bounded Variation)**.** *Every monotone function on $[a, b]$ has bounded variation.*

Professional Standard Proof.
Assume first that α is increasing. For every partition P,

$$\sum_i |\alpha(x_i) - \alpha(x_{i-1})| = \sum_i \big(\alpha(x_i) - \alpha(x_{i-1})\big) = \alpha(b) - \alpha(a),$$

because the middle sum telescopes. Hence

$$V_a^b(\alpha) = \alpha(b) - \alpha(a) < \infty.$$

If α is decreasing, then $-\alpha$ is increasing, so

$$V_a^b(\alpha) = V_a^b(-\alpha) = (-\alpha)(b) - (-\alpha)(a) = \alpha(a) - \alpha(b) < \infty.$$

□

Detailed Learning Proof.
A monotone function never reverses direction. For an increasing function every increment is nonnegative, so the absolute values do nothing and all interior terms cancel. For a decreasing function, negating it turns it into an increasing function without changing the sizes of the jumps. □

Remark (Proof structure). Use telescoping for increasing functions and reduce decreasing functions to the increasing case by negation.

Remark (Dependencies).

- Total Variation.
- Bounded Variation.

Remark (Return). Return to Theorem

Theorem (Continuous Integrand and BV Integrator)**.** *If f is continuous and α has bounded variation, then $\int f\,d\alpha$ exists.*

Professional Standard Proof.
Let $V = V_a^b(\alpha) < \infty$. Fix $\varepsilon > 0$. Since f is uniformly continuous on $[a, b]$, choose $\delta > 0$ such that every subinterval I of length less than δ satisfies

$$\omega_f(I) < \frac{\varepsilon}{3V + 1}.$$

Let (P, ξ) and (P', η) be tagged partitions with mesh less than δ, and let $Q = P \cup P'$ be a common refinement. First compare the Stieltjes sum on P to a compatible tagged sum on Q. On a slab I of P, refinement only replaces one tag value by tag values still lying in I. The total change over the pieces inside I is bounded by

$$\omega_f(I) \sum_{J \subset I} |\Delta\alpha(J)|.$$

Summing over all slabs of P gives

$$|S(f, \alpha, P, \xi) - S(f, \alpha, Q, \zeta)| \le \frac{\varepsilon}{3V + 1} \sum_J |\Delta\alpha(J)| < \frac{\varepsilon}{3}.$$

The same estimate compares the sum over P' to a compatible sum over Q. Finally, two taggings of the same refined partition Q differ by at most the same oscillation-times-variation estimate, hence by less than $\varepsilon/3$. The triangle inequality shows the original sums differ by less than ε. The sums form a Cauchy net in $\mathbb{R}$, so they converge. □

Detailed Learning Proof.
The ordinary Cauchy proof says "small oscillation times total width." The Stieltjes proof says "small oscillation times total α-variation." Uniform continuity supplies the small oscillation, while bounded variation supplies the finite total budget for $|\Delta\alpha|$. The common refinement is still the comparison device. □

Remark (Proof structure). Repeat the Cauchy common-refinement argument with α-increments controlled by total variation.

Remark (Dependencies).

- Riemann–Stieltjes Integral.
- Bounded Variation.
- Common Refinement of Two Partitions.

Remark (Return). Return to Theorem

Theorem (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist, the Riemann–Stieltjes integral is linear in the integrand and in the integrator.*

Professional Standard Proof.
TODO: professional standard proof for Riemann–Stieltjes bilinearity. □

Detailed Learning Proof.
TODO: detailed learning proof for Riemann–Stieltjes bilinearity. □

Remark (Proof structure).

Remark (Dependencies).

- Riemann–Stieltjes Integral

Remark (Return). Return to Theorem

Theorem (Interval Additivity for the Riemann–Stieltjes Integral). *When the relevant integrals exist, the Riemann–Stieltjes integral over $[a, b]$ splits across an intermediate point c.*

Professional Standard Proof.
TODO: professional standard proof for Riemann–Stieltjes interval additivity. □

Detailed Learning Proof.
TODO: detailed learning proof for Riemann–Stieltjes interval additivity. □

Remark (Proof structure).

Remark (Dependencies).

- Riemann–Stieltjes Integral

Remark (Return). Return to Theorem

Theorem (Riemann–Stieltjes Integration by Parts)**.** *Under the Riemann–Stieltjes convention, existence of one integral implies existence of the paired integral and their sum is the endpoint product difference.*

Professional Standard Proof.
TODO: professional standard proof for Riemann–Stieltjes integration by parts. ☐

Detailed Learning Proof.
TODO: detailed learning proof for Riemann–Stieltjes integration by parts. ☐

Remark (Proof structure).

Remark (Dependencies).

- Riemann–Stieltjes Integral

Remark (Return). Return to Theorem

Theorem (Reduction to Riemann Integration for a C^1 Integrator)**.** *If the integrator is C^1, then the Riemann–Stieltjes integral reduces to an ordinary Riemann integral against α'.*

Professional Standard Proof.
TODO: professional standard proof for the C^1 Riemann–Stieltjes reduction. □

Detailed Learning Proof.
TODO: detailed learning proof for the C^1 Riemann–Stieltjes reduction. □

Remark (Proof structure).

Remark (Dependencies).

- Riemann–Stieltjes Integral
- Riemann Integral

Remark (Return). Return to Theorem

Theorem (Step Integrators Give Finite Sums). *A Riemann–Stieltjes integral against a finite-step integrator is a finite weighted sum over the jump points, provided the integrand is continuous there.*

Professional Standard Proof.
TODO: professional standard proof for finite-step Riemann–Stieltjes integrators. ☐

Detailed Learning Proof.
TODO: detailed learning proof for finite-step Riemann–Stieltjes integrators. ☐

Remark (Proof structure).

Remark (Dependencies).

- Riemann–Stieltjes Integral

Remark (Return). Return to Lemma

Lemma (Cousin's Lemma). *For every gauge δ on $[a,b]$, there exists a δ-fine tagged partition of $[a,b]$.*

Professional Standard Proof.
Assume no δ-fine tagged partition exists. Bisect $[a,b]$. At least one half must also admit no δ-fine tagged partition, because partitions of both halves could be concatenated. Iterating gives nested closed intervals $[a_n, b_n]$ with lengths tending to 0, each admitting no δ-fine tagged partition. By the nested interval property their intersection is a single point c. Since $\delta(c) > 0$, some interval $[a_N, b_N]$ is contained in $(c - \delta(c), c + \delta(c))$. The one-slab partition of $[a_N, b_N]$, tagged at c, is then δ-fine, a contradiction. □

Detailed Learning Proof.
The only possible way a gauge could fail is if every attempted chopping has a bad subinterval. Repeated bisection traps those bad subintervals around one point. But the gauge is positive at that point, so sufficiently small trapped intervals fit inside its gauge ball and are valid after all. This contradiction proves non-vacuity. □

Remark (Proof structure). Assume failure, repeatedly bisect to produce nested bad intervals, then use compactness and positivity of the gauge at the limiting point to build the forbidden one-slab partition.

Remark (Dependencies).

- Gauge on an Interval.
- Delta-Fine Tagged Partition.

Remark (Return). Return to Theorem

Theorem (Riemann Integrable Implies Henstock–Kurzweil Integrable). *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, with the same value.*

Professional Standard Proof.
Let L be the Riemann integral of f. Given $\varepsilon > 0$, choose $\eta > 0$ so that every tagged partition with mesh less than η has Riemann sum within ε of L. Use the constant gauge $\delta(x) = \eta/2$. Every δ-fine tagged partition then has mesh less than η, so its tagged sum is within ε of L. □

Detailed Learning Proof.
The Henstock–Kurzweil definition allows more gauges than Riemann needs. To recover Riemann's situation, take a gauge that is constant at every point. A δ-fine partition for that constant gauge has every subinterval shorter than the Riemann mesh tolerance, so the old Riemann estimate applies without change. □

Remark (Proof structure). Embed the uniform mesh condition into the gauge condition by choosing a constant gauge.

Remark (Dependencies).

- Riemann Integral.
- Gauge on an Interval.
- Delta-Fine Tagged Partition.
- Henstock–Kurzweil Integral.

Remark (Return). Return to Lemma

Lemma (Straddle Lemma)**.** *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then the one-slab linearization error across any sufficiently small interval straddling ξ is bounded by ε times its length.*

Professional Standard Proof.
By differentiability, choose $\delta(\xi) > 0$ such that

$$|F(t) - F(\xi) - f(\xi)(t - \xi)| \le \varepsilon|t - \xi|$$

whenever $|t - \xi| < \delta(\xi)$. Apply this once at $t = v$ and once at $t = u$. Since

$$F(v) - F(u) = \big(F(v) - F(\xi)\big) - \big(F(u) - F(\xi)\big),$$

the triangle inequality gives

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - \xi) + \varepsilon(\xi - u) = \varepsilon(v - u).$$

□

Detailed Learning Proof.
Differentiability says that near ξ, the increment from ξ to any nearby point is well approximated by the linear term $f(\xi)(t - \xi)$. Apply that estimate to the right endpoint v and the left endpoint u. The errors add, and the distances $(v - \xi)$ and $(\xi - u)$ add to $v - u$. □

Remark (Proof structure). Apply the pointwise differentiability estimate at both ends of a tagged slab and subtract the two linear approximations.

Remark (Dependencies).

- Delta-Fine Tagged Partition.

Remark (Return). Return to Theorem

Theorem (Fundamental Theorem for the Henstock–Kurzweil Integral)**.** *If F is differentiable on $[a, b]$ and $F' = f$, then*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Professional Standard Proof.
Fix $\varepsilon > 0$ and set $\varepsilon' = \varepsilon/(b - a)$. At each tag ξ, apply the Straddle Lemma with tolerance ε', producing a gauge $\delta(\xi) > 0$. For any δ-fine tagged partition,

$$|f(\xi_i)\Delta x_i - (F(x_i) - F(x_{i-1}))| \le \varepsilon' \Delta x_i$$

on every slab. Summing and using telescoping gives

$$|S(f, P, \xi) - (F(b) - F(a))| \le \sum_i \varepsilon' \Delta x_i = \varepsilon'(b - a) = \varepsilon.$$

Thus every δ-fine tagged sum is within ε of $F(b) - F(a)$. □

Detailed Learning Proof.
The gauge is built from the local differentiability estimate at each possible tag. Because each slab straddles its tag and lies inside the tag's gauge ball, the local linear approximation controls that slab. The F-increments telescope, so the whole Riemann sum is forced near $F(b) - F(a)$. □

Remark (Proof structure). Use differentiability to build a gauge, apply the Straddle Lemma on each gauge-fine slab, and telescope the resulting F-increments.

Remark (Dependencies).

- Cousin's Lemma.
- Straddle Lemma.
- Henstock–Kurzweil Integral.

Remark (Return). Return to Theorem

Theorem (Continuous Functions Are Henstock–Kurzweil Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

Professional Standard Proof.
A continuous function on a compact interval is Cauchy-integrable. Cauchy's tag independence theorem identifies the same value for all sufficiently fine tagged sums. Thus the function is Riemann-integrable. By the inclusion theorem, every Riemann-integrable function is Henstock–Kurzweil integrable with the same value. □

Detailed Learning Proof.
Continuity gives enough uniform control for the ordinary Riemann integral. Once the Riemann integral exists, the gauge definition does not create a new obstruction: a constant gauge reproduces the Riemann mesh condition. Thus every continuous function enters the Henstock–Kurzweil class automatically. □

Remark (Proof structure). Use the existing continuous-implies-Riemann theorem, then pass through the Riemann-to-Henstock–Kurzweil inclusion.

Remark (Dependencies).

- Continuous Functions Are Cauchy-Integrable.
- Tag Independence for Continuous Functions.
- Riemann Integrable Implies Henstock–Kurzweil Integrable.

Remark (Return). Return to Theorem

Theorem (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a,b] \subseteq \mathcal{M}[a,b] \subseteq \mathcal{HK}[a,b],$$

with agreement of integral values.

Professional Standard Proof.
First suppose f is McShane integrable with value A. Every Henstock–Kurzweil δ-fine tagged partition is McShane δ-fine, because HK adds the extra restriction that each tag lies in its own interval. Hence the McShane estimate applies to all HK partitions, so f is HK-integrable with the same value.

Now suppose f is Riemann integrable with value L. By the Lebesgue criterion, its discontinuity set has measure zero. Let M bound $|f|$. Choose an open cover G of the discontinuity set with total length so small that $2M|G| < \varepsilon/3$. Define the gauge at discontinuity points so that their gauge balls lie inside G. At each continuity point x, choose the gauge so small that $|f(y) - f(x)|$ is small whenever y lies in the gauge ball.

For any McShane-fine partition, split the intervals according to whether their tags are discontinuity points or continuity points. The first class has total length controlled by $|G|$, hence contributes less than $\varepsilon/3$. On the second class, the free tag may sit outside its interval, but the entire interval lies inside the tag's continuity ball, so the oscillation error is small. Comparing these sums with ordinary Darboux sums for f and using the Riemann value L gives

$$|S(f, P, \xi) - L| < \varepsilon.$$

Thus f is McShane integrable and the value is L. □

Detailed Learning Proof.
The inclusion $\mathcal{M} \subseteq \mathcal{HK}$ is a quantifier argument: McShane tests more partitions. If a function survives every McShane-fine partition, it automatically survives the smaller family of HK-fine partitions.

The inclusion $\mathcal{R} \subseteq \mathcal{M}$ is the same philosophy as the Lebesgue criterion. Riemann integrability says the bad points are null. Around those points the gauge is made so small that all intervals tagged there occupy negligible total length. Away from those points the function is continuous, so a tag controls every nearby point in the interval it authorizes. Free tags do not break the argument because the McShane condition still forces the whole interval to lie inside the tag's gauge ball. □

Remark (Proof structure). Use the inclusion of partition classes for McShane ⇒ HK, and use the null discontinuity set plus local continuity gauges for Riemann ⇒ McShane.

Remark (Forward dependency). This proof uses the Lebesgue criterion proved in the next proof topic. The chapter order presents McShane before measure zero for conceptual flow, but the dependency graph points forward to the criterion.

Remark (Dependencies).

- McShane-Fine Tagged Partition.
- McShane Integral.
- Henstock–Kurzweil Integral.
- Lebesgue Criterion for Riemann Integrability.

Remark (Return). Return to Theorem

Theorem (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a,b] = \mathcal{L}[a,b],$$

with agreement of integral values.

Professional Standard Proof.
Show first that a Lebesgue-integrable function admits gauges for which every McShane-fine sum is close to the Lebesgue integral. Approximate by simple functions, control the exceptional level-set boundaries by small total measure, and use free tags to force the same absolute-size bookkeeping as the Lebesgue construction.

Conversely, show that McShane integrability implies Lebesgue integrability by testing the free-tag sums against positive and negative parts. The freedom to tag intervals from nearby points rules out the conditional cancellations that HK permits, so the resulting indefinite integral is absolutely continuous and has f as its derivative almost everywhere. □

Detailed Learning Proof.
The proof has two conceptual halves. Lebesgue integrability gives enough measure control to make all free-tag gauge sums stable. McShane integrability is strong enough to recover that same measure control, because the tags are not tethered to the cells where cancellation would otherwise hide large positive and negative oscillations. □

Remark (Proof structure). Reduce Lebesgue ⇒ McShane to simple-function approximation and small exceptional sets. Reduce McShane ⇒ Lebesgue to absolute continuity of the associated indefinite integral.

Remark (Dependencies).

- McShane-Fine Tagged Partition.
- McShane Integral.
- Lebesgue Criterion for Riemann Integrability.

Remark (Return). Return to Theorem

Theorem (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set has measure zero.*

Professional Standard Proof.
Let

$$\Omega = \sup_{[a,b]} f - \inf_{[a,b]} f < \infty.$$

First suppose the discontinuity set D has measure zero. Fix $\varepsilon > 0$. Choose $\eta > 0$ with $\eta(b - a) < \varepsilon/2$. Let

$$D_\eta = \{x : \omega_f(x) \geq \eta\}.$$

Then $D_\eta \subseteq D$, so D_η has measure zero. Cover D_η by open intervals whose total length is less than $\varepsilon/(2\Omega)$; by compactness of D_η, take a finite subcover and call its union G. On $K = [a, b] \setminus G$, every point has oscillation $< \eta$. Compactness of K gives a single scale $\delta > 0$ such that every interval of length less than δ meeting K has oscillation $< \eta$.

Choose a partition P with mesh $< \delta$ that includes the endpoints of the intervals making up G. Slabs contained in G have total width less than $\varepsilon/(2\Omega)$, so their contribution to $U(f, P) - L(f, P)$ is less than $\varepsilon/2$. Slabs meeting K have oscillation $< \eta$, so their total contribution is less than $\eta(b - a) < \varepsilon/2$. Hence

$$U(f, P) - L(f, P) < \varepsilon.$$

By the Darboux criterion, f is Riemann-integrable.

Conversely, suppose D does not have measure zero. Since

$$D = \bigcup_{k=1}^{\infty} D_{1/k},$$

some D_{η_0} with $\eta_0 = 1/k_0$ does not have measure zero. Thus there is $c > 0$ such that every interval cover of D_{η_0} has total length at least c. For any partition P, the slabs whose interiors meet D_{η_0}, together with the finitely many partition points, cover D_{η_0}; ignoring finitely many points does not affect the lower bound on total interval length. Hence those slabs have total width at least c. Each such slab has oscillation at least η_0, so

$$U(f, P) - L(f, P) = \sum_i \omega_f([x_{i-1}, x_i])\Delta x_i \geq \eta_0 c > 0.$$

The Darboux gap cannot be made arbitrarily small, so f is not integrable. □

Detailed Learning Proof.
The proof says that all Darboux error lives over points where the function oscillates. If those bad points fit inside intervals of arbitrarily small total length, their worst possible contribution is small, and the good complement has uniformly small oscillation. If the bad set has positive unavoidable length at some oscillation level, every partition pays a fixed positive Darboux gap. □

Remark (Proof structure). Use the sets $D_\eta = \{x : \omega_f(x) \geq \eta\}$ to separate large oscillation from small oscillation, then apply the Darboux criterion in both directions.

Remark (Dependencies).

- Measure Zero.
- Point Oscillation.
- Darboux Criterion.

Bibliography